

Beyond Fake or Real: Evidence-Grounded, Explainable Five-Class News Verification with Temporal Context Checking

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Abstract—Most automatic “fake news” detectors are binary text classifiers: they learn how fabricated articles tend to be written rather than whether the claims in them are true, so they cannot explain a verdict, cannot describe an item that is only partly wrong or true-but-out-of-date, and cannot admit that they do not know. We present a compact, fully open-source verification pipeline that can run on a student laptop. Given a headline, an article or a URL, the system extracts check-worthy claims, retrieves evidence from free online sources and from an offline index of fact-checked statements, labels the stance of every evidence passage with a natural-language-inference cross-encoder, compares publication dates to detect old news recirculated as “breaking”, and fuses these signals with transparent rules into one of five verdicts—Real, Fake, Partially True, Misleading or Unverifiable—together with a confidence score, dated evidence cards and a plain-language explanation. A fine-tuned DistilBERT content classifier with LIME token attributions contributes only to the confidence estimate. On the WELFake corpus the classifier reaches a macro-F1 of [TODO: 0.xx], and on a hand-labelled set of about 100 recent claims the complete pipeline reaches a five-class macro-F1 of [TODO: 0.xx]; a cross-dataset audit shows why in-domain accuracies near 99 % do not transfer. Code, data splits and the claim set are released.

Index Terms—fake news detection, fact verification, evidence retrieval, stance detection, natural language inference, explainable AI, temporal misinformation

I. Introduction

News now reaches most people through feeds and messaging groups rather than through editors, and the damage done by what slips through is well documented. Tracing roughly 126,000 story cascades on Twitter over eleven years, Vosoughi et al. found that false stories were about 70 % more likely to be retweeted than true ones and that the truth needed roughly six times longer to reach 1,500 people [1]. The World Economic Forum’s Global Risks Report 2024 places misinformation and disinformation first among global risks for the coming two years, ahead of extreme weather and armed conflict [2]. A study of the COVID-19 “infodemic” linked a single rumour—that drinking concentrated alcohol or methanol cures the virus—to about 800 deaths and roughly 5,800

hospital admissions in the first three months of 2020 [3]. India, the setting of this work, offers its own examples: child-kidnapping rumours forwarded on WhatsApp were linked to at least 17 mob killings across the country between April and July 2018 [4], [5], and Indian fact-checkers routinely find genuine footage of earlier floods, riots and accidents re-captioned as “breaking” news for a new event—in one analysis of 125 fact-checked Indian COVID-19 items, nearly half paired text with video and 94 % originated on social platforms [6]. This last pattern—true material presented in the wrong time or place—is one that most detectors are not built to see.

Manual fact-checking is accurate but slow. Most academic work on automatic detection has therefore taken a shortcut: treat the task as binary text classification and let a model learn the surface cues of fabricated writing. Such classifiers report accuracies above 95 % on benchmark corpora [7], [8], but the numbers hide three weaknesses. First, they learn style and dataset artefacts rather than facts: a corpus whose genuine articles all come from one wire service can be separated almost perfectly by boilerplate alone [9], and the same models lose much of their accuracy on a different corpus. Second, a binary label cannot express the shapes real misinformation takes—a true event with a wrong number, a real quotation attached to the wrong occasion, a 2018 video captioned as today’s news. Third, a classifier always produces an answer; it cannot say that the evidence is insufficient, which for hyper-local or very recent claims is the only honest response.

Problem statement. Given a news item—a headline, an article text or a URL—automatically determine, with supporting evidence, dates and a human-readable explanation, whether its central claims are real, fake, partially true, misleading or unverifiable, using only free tools and hardware available to students.

We address this problem by moving the decision from the classifier to the evidence: the system selects the check-worthy sentences of an item, retrieves passages from free

web search, Wikipedia, the Google Fact Check Tools API and an offline index of fact-checked statements, labels each passage as supporting, refuting or neutral, compares the dates involved, and applies a small, inspectable set of rules. A style-based classifier is retained, but it may only adjust the confidence or break a single-source tie; it never outputs Fake or Real on its own when evidence exists. The contributions of this paper are:

- An evidence-grounded five-class verdict scheme with abstention. We define Real, Fake, Partially True, Misleading and Unverifiable operationally, map LIAR’s six PolitiFact grades onto them, and reach each verdict through a published decision table whose confidence reflects evidence agreement, coverage and source reliability, so every output carries the rule that fired, the evidence used and an explicit uncertainty statement (objective O6).
- A laptop-scale claim-to-verdict pipeline. Claim identification, evidence retrieval from free sources with an offline FAISS fallback, NLI stance detection with a small cross-encoder, and a temporal/context check that flags old news presented as new and entity-date mismatches, all running on CPU within seconds offline (objectives O3–O5).
- An explainable content classifier evaluated honestly. A fine-tuned DistilBERT model with LIME token attributions serves as the style signal; we report in-domain results on WELFake, a cross-dataset transfer and leakage audit against ISOT, and an ablation of how much the classifier adds to the evidence-based verdict (objective O1).

The rest of the paper is organised as follows. Section II surveys related work and states the gap; Sections III–V describe the datasets, the pipeline with its fusion rules, and the experimental setup; Section VI reports results, Section VII compares them with existing systems, and Section VIII concludes.

II. Related Work

We group prior work into five themes that correspond to the components of our system; Table I summarises the most relevant papers along the dimensions that matter to us.

A. Datasets and Benchmarks

LIAR [10] collects 12.8k short political statements rated by PolitiFact on a six-point scale from pants-on-fire to true. Its best text-and-metadata model reached only about 27 % six-way accuracy, which remains a useful reminder that fine-grained truthfulness cannot be read off the wording of a sentence. ISOT [9], [11] pairs 21k Reuters articles with 23k articles from sites flagged by fact-checkers; because every genuine article shares one source, the corpus is easy to separate and is widely suspected of source leakage. WELFake [7] merges four public corpora into 72k articles to reduce exactly this kind of overfitting

and is our primary training set. FakeNewsNet [12] adds social context and PolitiFact/GossipCop labels to news items; we use its PolitiFact titles in our offline index. On the verification side, FEVER [13] contributes 185k claims written against Wikipedia and labelled Supports, Refutes or Not Enough Info, defining the retrieve-then-verify task that our stance module inherits, while MultiFC [14] gathers 35k claims with the heterogeneous verdict labels of 26 real fact-checking sites. ClaimBuster [15] provides 23.5k debate sentences annotated for check-worthiness. Surveys by Zhou and Zafarani [16] and Guo et al. [17] organise this landscape: the former distinguishes knowledge-, style-, propagation- and source-based detection, and the latter formalises fact-checking as claim detection, evidence retrieval, verdict prediction and justification production—the sequence our pipeline follows.

B. Content-Based Classifiers

The earliest and still most common approach treats detection as document classification. Ahmed et al. [9] reported that TF-IDF n-grams with a linear SVM reach roughly 92 % accuracy on ISOT, and Pérez-Rosas et al. [18] found linear models on lexical, syntactic and readability features to be competitive on crowd-sourced fake news. Pre-trained transformers [19] raised these numbers further: FakeBERT [8] feeds BERT embeddings through parallel convolutional blocks and reports 98.9 % accuracy on a Kaggle corpus, and WELFake’s own linguistic-feature model reports 96.7 %. DistilBERT [20] keeps about 97 % of BERT’s language-understanding performance at 40 % fewer parameters, which is what makes fine-tuning on a 4 GB consumer GPU feasible and is why we adopt it. What these papers share is an in-domain, binary evaluation; none reports cross-corpus transfer, and none can tell a user why an article is suspicious beyond its wording. We keep such a classifier, but only as a secondary signal.

C. Claim Detection, Evidence Retrieval and Fact Verification

ClaimBuster [21] showed that check-worthy sentences in political debates can be ranked with sentence-level features and a supervised classifier, and its later benchmark release [15] is the standard training set for the task; our claim identifier combines similar hand-crafted cues with an optional classifier trained on that data. For verification, the FEVER baseline [13] chains document retrieval, sentence selection and an NLI model and scored 31.9 FEVER points; the shared-task winner, NSMN [22], unified the three steps with semantic matching networks, and KGAT [23] propagates evidence through a kernel-based graph attention network to exceed 70 FEVER points—the current ceiling for Wikipedia-grounded verification. DeClarE [24] verifies free-text claims against web search results with attention over the retrieved articles and a source-credibility embedding, which is close in spirit to our online path. Two components make retrieval and

stance practical at our scale: Sentence-BERT [25] bi-encoders (we use the 22M-parameter MiniLM variant [26]) turn semantic search over thousands of fact-checked statements into a millisecond nearest-neighbour lookup, while a small DeBERTaV3 cross-encoder [27] trained on MultiNLI [28] and FEVER-NLI provides an off-the-shelf three-way stance judgement on CPU. Hanselowski et al. [29] re-examined the Fake News Challenge and showed that headline-body stance systems that looked strong under the challenge metric had far lower macro-F1 on the minority agree/disagree classes—a caution we take into account by reporting macro-F1 throughout.

D. Explainability

LIME [30] explains an individual prediction by fitting a sparse linear surrogate on perturbed inputs, and SHAP [31] unifies several attribution methods under Shapley values; both are model-agnostic and produce token-level highlights that lay users can read. In the fake-news setting, dEFEND [32] co-attends over article sentences and user comments so that the sentences and comments that drove the decision can be shown as the explanation, and DeClarE [24] exposes the retrieved evidence words its attention focused on. These methods explain a model; they do not by themselves ground the verdict in checkable facts. We therefore treat LIME highlights as an explanation of the style signal only, and build the primary explanation from the retrieved evidence, its stance, its source and its date.

E. Temporal and Contextual Misinformation

Much misinformation is not fabricated text but genuine material displaced in time or place. Vosoughi et al. [1] quantified how much faster and deeper falsehoods travel, and Zubiaga et al. [33] survey rumour detection, tracking and stance classification as an inherently temporal process in which a story’s veracity is resolved over time. In India, Al-Zaman [6] analysed 125 fact-checked COVID-19 stories and found that two-thirds concerned health, that text paired with video was the commonest format, and that almost all originated on social platforms rather than in mainstream media; reporting on the 2018 WhatsApp lynchings [4], [5] traced the killings to rumour messages and videos forwarded as local warnings. Existing detectors ignore publication dates entirely; we add a lightweight check that compares the input’s date and recency cues against the dates of the supporting evidence and against the entities it mentions.

F. The Gap

Table I makes the gap visible. Style-based classifiers are accurate in-domain but produce no evidence, no partial-truth labels and no abstention; FEVER-style verifiers ground their decisions in evidence but assume Wikipedia-only, synthetic claims and do not consider time; explainability work explains models rather than facts. No system

TABLE I

Representative prior work compared on the properties this paper targets. Ev. = grounds the verdict in retrieved evidence; Exp. = explains the verdict to a user; Abst. = can abstain; Temp. = checks dates or context. Full table in docs/literature_table.md.

Work	Output	Ev.	Exp.	Abst.	Temp.
LIAR CNN [10]	6-way	—	—	—	—
ISOT SVM [9]	binary	—	—	—	—
WELFake [7]	binary	—	—	—	—
FakeBERT [8]	binary	—	—	—	—
FEVER baseline [13]	3-way	✓	—	NEI	—
KGAT [23]	3-way	✓	—	NEI	—
DeClarE [24]	binary	✓	✓	—	—
dEFEND [32]	binary	—	✓	—	—
MultiFC [14]	per-site	✓	—	—	—
This work	5-way	✓	✓	✓	✓

we are aware of combines (i) a five-way verdict with an explicit Unverifiable class, (ii) evidence retrieval from free sources with an offline fallback, (iii) a temporal and contextual check, and (iv) evidence-linked explanations, while running on ordinary student hardware. Those properties are the three contributions listed in Section I, and the rest of this paper builds and evaluates them.

III. Datasets and Preprocessing

Placeholder — to be written for MSE2.

IV. Methodology

Placeholder — to be written for MSE2.

V. Experimental Setup

Placeholder — to be written for MSE2.

VI. Results

Placeholder — to be written for MSE2 (partial) and ESE (final).

VII. Discussion and Comparison with Existing Systems

Placeholder — to be written for ESE.

VIII. Conclusion and Future Work

Placeholder — to be written for ESE.

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References

- [1] S. Vosoughi, D. Roy, and S. Aral, “The spread of true and false news online,” *Science*, vol. 359, no. 6380, pp. 1146–1151, 2018.
- [2] World Economic Forum, “The global risks report 2024,” World Economic Forum, Geneva, Switzerland, Tech. Rep., Jan. 2024, 19th edition. <https://www.weforum.org/publications/global-risks-report-2024/>.
- [3] M. S. Islam, T. Sarkar, S. H. Khan, A.-H. Mostofa Kamal, S. M. M. Hasan, A. Kabir, D. Yeasmin, M. A. Islam, K. I. Amin Chowdhury, K. S. Anwar, A. A. Chughtai, and H. Seale, “COVID-19-related infodemic and its impact on public health: A global social media analysis,” *The American Journal of Tropical Medicine and Hygiene*, vol. 103, no. 4, pp. 1621–1629, 2020.
- [4] D. Bathini, “How WhatsApp helped turn an Indian village into a lynch mob,” *BBC News*, <https://www.bbc.com/news/world-asia-india-44856910>, Jul. 2018, 18 July 2018; accessed 28 Aug 2026.
- [5] Al Jazeera, “In India, WhatsApp stirs up deadly rumours,” <https://www.aljazeera.com/features/2018/7/17/in-india-whatsapp-stirs-up-deadly-rumours>, Jul. 2018, 17 July 2018; accessed 28 Aug 2026.
- [6] M. S. Al-Zaman, “COVID-19-related social media fake news in India,” *Journalism and Media*, vol. 2, no. 1, pp. 100–114, 2021.
- [7] P. K. Verma, P. Agrawal, I. Amorim, and R. Prodan, “WELFake: Word embedding over linguistic features for fake news detection,” *IEEE Transactions on Computational Social Systems*, vol. 8, no. 4, pp. 881–893, 2021.
- [8] R. K. Kaliyar, A. Goswami, and P. Narang, “FakeBERT: Fake news detection in social media with a BERT-based deep learning approach,” *Multimedia Tools and Applications*, vol. 80, pp. 11 765–11 788, 2021.
- [9] H. Ahmed, I. Traore, and S. Saad, “Detection of online fake news using N-gram analysis and machine learning techniques,” in *Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments (ISDDC 2017)*, ser. Lecture Notes in Computer Science, vol. 10618. Springer, 2017, pp. 127–138.
- [10] W. Y. Wang, ““Liar, Liar Pants on Fire”: A new benchmark dataset for fake news detection,” in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Vancouver, Canada: Association for Computational Linguistics, 2017, pp. 422–426.
- [11] H. Ahmed, I. Traore, and S. Saad, “Detecting opinion spams and fake news using text classification,” *Security and Privacy*, vol. 1, no. 1, p. e9, 2018.
- [12] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, “FakeNewsNet: A data repository with news content, social context, and spatiotemporal information for studying fake news on social media,” *Big Data*, vol. 8, no. 3, pp. 171–188, 2020.
- [13] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, “FEVER: A large-scale dataset for fact extraction and VERification,” in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. New Orleans, Louisiana: Association for Computational Linguistics, 2018, pp. 809–819.
- [14] I. Augenstein, C. Lioma, D. Wang, L. Chaves Lima, C. Hansen, C. Hansen, and J. G. Simonsen, “MultiFC: A real-world multi-domain dataset for evidence-based fact checking of claims,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 4685–4697.
- [15] F. Arslan, N. Hassan, C. Li, and M. Tremayne, “A benchmark dataset of check-worthy factual claims,” in *Proceedings of the Fourteenth International AAAI Conference on Web and Social Media (ICWSM)*, vol. 14, 2020, pp. 821–829.
- [16] X. Zhou and R. Zafarani, “A survey of fake news: Fundamental theories, detection methods, and opportunities,” *ACM Computing Surveys*, vol. 53, no. 5, pp. 109:1–109:40, 2020.
- [17] Z. Guo, M. Schlichtkrull, and A. Vlachos, “A survey on automated fact-checking,” *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 178–206, 2022.
- [18] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, “Automatic detection of fake news,” in *Proceedings of the 27th International Conference on Computational Linguistics (COLING)*. Santa Fe, New Mexico, USA: Association for Computational Linguistics, 2018, pp. 3391–3401.
- [19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186.
- [20] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019, presented at the 5th Workshop on Energy Efficient Machine Learning and Cognitive Computing (NeurIPS 2019).
- [21] N. Hassan, F. Arslan, C. Li, and M. Tremayne, “Toward automated fact-checking: Detecting check-worthy factual claims by ClaimBuster,” in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Halifax, NS, Canada: ACM, 2017, pp. 1803–1812.
- [22] Y. Nie, H. Chen, and M. Bansal, “Combining fact extraction and verification with neural semantic matching networks,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, no. 01, 2019, pp. 6859–6866.
- [23] Z. Liu, C. Xiong, M. Sun, and Z. Liu, “Fine-grained fact verification with kernel graph attention network,” in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 2020, pp. 7342–7351.
- [24] K. Popat, S. Mukherjee, A. Yates, and G. Weikum, “DeClarE: Debunking fake news and false claims using evidence-aware deep learning,” in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 22–32.
- [25] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992.
- [26] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5776–5788.
- [27] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- [28] A. Williams, N. Nangia, and S. Bowman, “A broad-coverage challenge corpus for sentence understanding through inference,” in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. New Orleans, Louisiana: Association for Computational Linguistics, 2018, pp. 1112–1122.
- [29] A. Hanselowski, A. PVS, B. Schiller, F. Caspelherr, D. Chaudhuri, C. M. Meyer, and I. Gurevych, “A retrospective analysis of the fake news challenge stance-detection task,” in *Proceedings of the 27th International Conference on Computational Linguistics (COLING)*. Santa Fe, New Mexico, USA: Association for Computational Linguistics, 2018, pp. 1859–1874.
- [30] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why should I trust you?”: Explaining the predictions of any classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. San Francisco, CA, USA: ACM, 2016, pp. 1135–1144.
- [31] S. M. Lundberg and S.-I. Lee, “A unified approach to inter-

- preting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.
- [32] K. Shu, L. Cui, S. Wang, D. Lee, and H. Liu, “dEFEND: Explainable fake news detection,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. Anchorage, AK, USA: ACM, 2019, pp. 395–405.
- [33] A. Zubiaga, A. Aker, K. Bontcheva, M. Liakata, and R. Procter, “Detection and resolution of rumours in social media: A survey,” *ACM Computing Surveys*, vol. 51, no. 2, pp. 32:1–32:36, 2018.